

Evan Nelles Henderson | Curriculum Vitae

MSc Student in Physics and Astronomy · McMaster University

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Research Interests

Origins of life on rocky exoplanets, atmospheric chemistry and climate evolution, planetary interiors and magma ocean processes, interior-atmosphere interactions.

Education

M.Sc. in Physics (Astrobiology Collaborative Program) 2025 – Present
McMaster University

Thesis: *Methane from the Mantle: Geophysical Pathways to Prebiotic Atmospheres (In Progress)*

Supervisor: Dr. Ralph Pudritz

B.Sc. in Physics (Minor Natural History) 2020 – 2024
McGill University

Research Experience & Fieldwork

Graduate Researcher — Planetary Interiors and Atmospheres 2025 – Present
McMaster University

Advisors: Dr. Ralph Pudritz & Dr. Tim Lichtenberg

- Modelling the coupled evolution of magma oceans and protoatmospheres on rocky planets using the PROTEUS framework, with the goal of identifying geophysical pathways to prebiotic atmospheres.
- Conducting parameter sweeps of volatile outgassing scenarios to characterise atmospheric composition during and after magma ocean solidification.
- Researching surface-atmosphere interactions, including volcanism, serpentinization, and carbonate formation.

A Bayesian Investigation of the Transit Method 2023
McGill University

Advisor: Dr. Adrian Liu

- Modelled exoplanet transit light curves using MCMC methods as part of a supervised undergraduate project.

Palaeontology Field Research 2023

Dinosaur Provincial Park, Alberta

Advisor: Dr. Hans Larsson

- Three weeks of fossil excavation and field data collection in the badlands as part of a supervised field research course.

Science Communication Experience

Author and Editor 2026 – Present
[Astrobites](#)

Science Communication Publications

- [Hang On To Your Water, Super-Earths!](#), *Astrobites*, Feb 17th, 2026.
- [Admiring the Atmospheric Egg Shell of Sub-Neptune TOI-270d](#), *Astrobites*, Apr 6th, 2026.

Talks & Posters

- Talk: “Magma Ocean-Protoatmosphere Evolution”, Astronomy Journal Club, McMaster University, 2026.
- Talk: “Mantle-Atmosphere Coupling on Terrestrial Planets”, McMaster-Waterloo Exoplanet Symposium, McMaster University, 2025.

Teaching Experience

Teaching Assistant 2025 – Present
McMaster University

- Integrated Science ([ISCI 1A24 A](#)) Fall 2025
- Introductory Mechanics ([PHYS 1D03](#)) Fall 2025
- Explorations in Medical and Biological Physics ([BIOPHYS 2S03](#)) Winter 2026
- Integrated Science ([ISCI 1A24 B](#)) Winter 2026

Technical Skills

Languages: Python, MATLAB, R, Bash, HTML, $\text{\LaTeX}$, NetLogo

Scientific: NumPy, Matplotlib, Jupyter, emcee, batman, PROTEUS (SPIDER, AGNI, FastChem)

Simulation: Climate modelling, prebiotic chemistry, transit light curves, protocell replication

Operational: Git, Linux, SSH, HPC, Anaconda

Awards & Scholarships

- McMaster Graduate Entrance Scholarship 2025
- Martlet Student Athlete Honour Roll (McGill University Varsity Rowing Team) 2023
- Scotiabank Scholarship (Scholarships Partners Canada) 2020 – 2024

References

Available upon request.